

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. CHB-501: Analytical Chemistry-I

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following:

[5 × 2 = 10]

- (a) What is the role of a masking agent? Give one example.
 - (b) Define systematic error with a suitable example.
 - (c) Define confidence interval.
 - (d) Write the chemical structure of oxine (8-hydroxyquinoline).
 - (e) Differentiate between iodometry and iodimetry.
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Question 2.

- (a) State the principles of gravimetric analysis. Describe the various steps involved in gravimetric analysis. [8]
 - (b) Discuss co-precipitation and post-precipitation, highlighting their differences. [6]
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Question 3.

- (a) Discuss various methods used to minimize errors in quantitative analysis. [6]
 - (b) The percentage of copper in an alloy was determined as: 12.35%, 12.42%, 12.39%, 12.31%, 12.44%. Calculate: (i) mean, (ii) standard deviation, and (iii) 95% confidence limit. (Given: $t = 2.78$ for 4 degrees of freedom). [8]
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Question 4.

- (a) Name the titration you would perform for the evaluation of organic functional groups in the following reactions: [8]
 - (i) $\text{R-OH} + \text{Ac}_2\text{O} \rightarrow \text{R-OAc} + \text{AcOH}$
 - (ii) $\text{R-CHO} + \text{NH}_2\text{OH} \cdot \text{HCl} \rightarrow \text{R-CH=NOH} + \text{H}_2\text{O} + \text{HCl}$
 - (b) Discuss the chemical principle, preparation, and standardization of Karl Fischer reagent for the estimation of moisture content. [6]
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Question 5. Discuss the theoretical and practical aspects of the following reagents in chemical analysis: [7 + 7 = 14]

- (a) EDTA in complexometric titrations
- (b) Potassium bromate (KBrO_3) in redox titrations